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Hair and artifact removal in dermoscopic images using deep learning for enhanced skin cancer detection

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The timely identification of melanoma and other cutaneous malignancies is fundamentally reliant on the acquisition of high-caliber dermoscopic images. However, factors such as hair interference, reflections, and various illumination artifacts frequently obscure essential characteristics of lesions, thereby impeding both clinical evaluations and automated diagnostic processes. This research introduces an innovative deep learning framework based on U-Net architecture, meticulously designed for the effective removal of hair and artifacts from dermoscopic images while ensuring the preservation of intricate lesion attributes, including pigmentary networks and irregular edges. The model underwent training predominantly on the HAM10000 dataset, which comprises 10,015 images sourced from multiple origins, and was rigorously validated on separate portions of both the HAM10000 dataset and the ISIC 2018 dataset. The training employed a hybrid loss function of Dice and Binary Cross-Entropy, alongside extensive data augmentation techniques and meticulous artifact mask creation. Quantitative assessments reveal significant improvements in image quality: Peak Signal-to-Noise Ratio (PSNR) increased from 21.5 dB to 34.1 dB, and Structural Similarity Index Measure (SSIM) enhanced from 0.79 to 0.89 for HAM10000 and from 0.77 to 0.92 for ISIC 2018, with Intersection over Union (IoU) ranging between 0.85 and 0.87 across the datasets. Subsequent melanoma classification utilizing a pre-trained model demonstrated notable improvements, with accuracy advancing from 84.2% to between 90.3% and 91.5%, F1-score rising from 81.6% to between 90.2% and 91.5%, along with an increase in prediction confidence. This methodology shows robust generalization capabilities across diverse artifact densities, types of lesions, and imaging environments, thereby establishing itself as a potent and adaptable preprocessing technique for standardized dermoscopic examinations and dependable AI-supported skin cancer diagnostics.

Keywords Public health, U-Net, Dermoscopic images, Cancer, Deep leaning, Skin cancer

Despite having a very low frequency among skin cancers, melanoma is the most deadly type, making it a significant worldwide health concern[1]. Global melanoma cases are steadily increasing, according to recent epidemiological statistics, underscoring the critical need for accurate and timely detection methods[2]. While delayed identification frequently results in poor clinical outcomes, early-stage melanoma is linked to a much greater survival rate. Extensive research on computer-aided and image-based diagnosis systems has been motivated by the substantial correlation between patient prognosis and diagnostic timing.

In recent years, the diagnosis of cancer has been greatly impacted by developments in artificial intelligence and medical imaging. Improved visualization of delicate anatomical structures is made possible by deep learning algorithms, which have shown great performance in image denoising and restoration tasks across several medical imaging modalities[3]. Simultaneously, traditional image enhancement techniques like contrast normalization and dehazing have been investigated to enhance image clarity in challenging acquisition scenarios[4]. The usefulness

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of data-driven pixel-level modeling has been demonstrated in more recent years by the successful application of learning-based segmentation techniques in challenging clinical situations, such as minimally invasive surgery[5]. Image quality is still a major limiting factor in diagnostic reliability in spite of these advancements.

Automated analysis systems are still challenged by low-quality medical images, especially those impacted by weak contrast and low illumination. A number of improvement methods based on logarithmic transformations and perceptual models have been proposed to overcome these problems and increase visibility in low light[6]. These techniques, however, are often made for natural photographs and do not specifically take into account the morphological and chromatic features of skin diseases. Many machine learning techniques have been developed in dermatology to classify skin lesions using ensemble learning frameworks, statistical descriptors, and custom features[7]. Despite the encouraging outcomes of these methods, image artifacts frequently weaken their resilience.

By enabling high-resolution reconstruction and super-resolution imaging, deep learning has further improved medical imaging by recovering finer structural details from noisy inputs[8]. Recent research has concentrated on improving lesion classification in dermoscopy by utilizing geometric features, superpixel representations, and improved feature fusion techniques[9]. across order to increase the accuracy of skin cancer staging and classification across extensive healthcare systems, several studies have investigated hybrid models and decision-support frameworks[10]. To improve lesion discrimination, morphological and color-based feature extraction techniques have also been studied[11]. However, these studies usually presume that there are no substantial artifacts in dermoscopic images previously.

Acquisition noise, poor contrast, and inconsistent illumination commonly deteriorate dermoscopic images in actual clinical settings. To counteract such degradations, adaptive improvement methods based on color compensation and gamma correction have been proposed[12]. Convolutional neural networks, which learn hierarchical representations straight from dermoscopic pictures, have been used more recently to enhance lesion classification ability[13]. These models' performance is nevertheless extremely vulnerable to image distortions that mask lesion borders and textures, despite their tremendous discriminative strength.

The non-invasive imaging method known as dermoscopy is well-established and allows for the observation of subsurface skin structures that are essential for melanoma assessment[14]. In order to precisely delineate lesion sites, fully convolutional networks have been presented for dermoscopic image segmentation, building on this modality[15]. The potential of end-to-end learning pipelines for diagnostic applications has been further supported by the success of deep learning-based detection frameworks in other medical imaging domains[16]. Nonetheless, hair and other visual distortions continue to have a significant impact on segmentation and classification performance.

Conventional image processing methods, like region growth and rule-based hair removal procedures, were used in early attempts to address artifact contamination in dermoscopic pictures[17]. Later, better optimization-based techniques were presented to improve lesion area extraction in difficult circumstances[18]. Despite these attempts, manual preprocessing techniques are not flexible enough and frequently fall short when faced with a variety of artifact patterns. This shortcoming has been further brought to light by the availability of huge, publicly available dermoscopic datasets, since real-world data show significant variation in acquisition quality and artifact characteristics[19,20].

When considered collectively, the body of extant material indicates a glaring vacuum. Even while lesion classification, segmentation, and enhancement have advanced significantly, artifact removal is frequently viewed as a secondary preprocessing step rather than a primary learning goal. There is a serious flaw in automated diagnostic processes since current approaches either presume clean input photos or rely on strict, handwritten restrictions. By putting forth a deep learning-based artifact removal framework created especially for dermoscopic images, this study immediately fills this gap. The suggested strategy is to enhance image quality in a manner that is directly compatible with subsequent diagnostic tasks by concentrating on pixel-level detection and suppression of hair and acquisition-related artifacts. The key methodological contributions of this work are summarized as follows:

- Instead of treating hair and acquisition-related artifacts as incidental noise, a specialized deep learning architecture for dermoscopic artifact removal is put forth.
- To separate artifacts from lesion structures, a pixel-level segmentation technique based on U-Net is utilized, which allows for efficient artifact suppression while maintaining clinically significant information.
- The resilience and dependability of automated melanoma detection pipelines are enhanced by artifact-aware preprocessing, according to thorough testing on extensive dermoscopic datasets.

The rest of the paper is organized as follows: In Sect. 2, a systematic review of classical and deep learning-based methods for hair and artifact removal is presented and their strengths and limitations are analyzed. In Sect. 3, Materials and Methods, the proposed U-Net-based framework is explained, data, preprocessing steps, model architecture, and evaluation criteria are described. Results and evaluation are reported in Sect. 4. Sections 5 and 6 summarize the discussion and conclusions, clinical implications, and future research directions.

Related works

Because it directly affects the accuracy of automated skin lesion analysis, the issue of hair and artifact contamination in dermoscopic pictures has garnered more attention. Learning-driven hair suppression was the main preprocessing step in early deep learning-based attempts to increase the accuracy of downstream diagnostics[21]. These techniques showed that convolutional architectures can perform better than traditional filtering techniques, but when faced with complex or varied artifact patterns, their effectiveness frequently deteriorates. In order to address data imbalance and class separability, parallel work in skin lesion analysis focused on representation learning and clustering techniques, but implicitly assumed artifact-free inputs[22].

Later research investigated hybrid approaches for automatic hair removal that combined deep neural networks and shallow learning[23]. Although these frameworks were more durable than solely handcrafted methods, their generalization potential was hampered by their limited architectural depth and frequent reliance on dataset-specific assumptions. Although related research in adaptive margin learning and content-based medical image retrieval significantly enhanced lesion class detection, it ignored the artifact removal issue and treated image quality as a fixed need rather than an adaptable pipeline component[24].

In order to explicitly describe hair patterns and contextual information in dermoscopic images, attention-based architectures were included in more recent research[25]. The preservation of lesion structures and artifact localization were greatly improved by dual-encoder designs and attention methods. Notwithstanding their efficacy, these models' sensitivity to the distribution of training data and increased computing complexity limited their scalability on big, diverse datasets. Concurrently, enhanced convolutional neural networks with a focus on border correctness and structural consistency were suggested for lesion segmentation[26]. However, these techniques did not specifically address artifact suppression; instead, they focused on lesion delineation.

In order to diagnose skin diseases, a number of studies combined deep learning and conventional machine learning methods, using a more comprehensive diagnostic approach[27]. Despite the fact that these methods enhanced classification performance, artifact handling was frequently done informally or through traditional preprocessing procedures. Later, more specialized deep learning models that performed well on specific datasets were presented for lesion analysis and joint hair artifact removal in developing dermatological applications[28]. These techniques, however, tended to concentrate on particular illnesses or imaging circumstances, which limited their generalizability.

Feature engineering, thresholding, and rule-based segmentation techniques were the mainstays of earlier classical methods for melanoma detection[29]. Although fundamental, these approaches were extremely susceptible to noise and artifacts, which prompted the creation of more reliable preprocessing methods. Although multi-filter heuristic methods that could eliminate hair and ruler markings were a significant advancement, their resilience in a variety of imaging conditions was diminished by their dependence on manually adjusted parameters[30].

The development of this discipline was greatly impacted by the availability of extensive public benchmarks. The heterogeneity of real-world dermoscopic pictures was brought to light and the systematic evaluation of lesion analysis techniques was made possible by standardized tasks and datasets[31]. Accelerated methodological development and reproducible experimentation were made possible by supporting software libraries for deep learning, image processing, and numerical computing[32–34]. But rather than solving the artifact problem directly, these tools offer the framework for algorithmic fixes.

Usually, objective picture quality and segmentation criteria are used to evaluate artifact removal techniques. Although PSNR's perceptual validity is restricted when used alone, it is frequently used to evaluate pixel-level quality[35]. While overlap-based metrics like IoU are still the gold standard for assessing segmentation accuracy[36], structural similarity metrics have gained popularity as a way to better capture perceptual and morphological preservation[37]. Persistent limitations in generalization and robustness across approaches are still evident in large, diversified datasets with real acquisition problems[20].

Overall, the literature shows a glaring gap: despite the fact that lesion classification, segmentation, and representation learning have advanced significantly, artifact removal is often viewed as an auxiliary preprocessing step rather than a primary learning goal. The scalability, generalization, and preservation of fine lesion morphology under various artifact settings remain unresolved issues with existing approaches, which either rely on computationally demanding designs, dataset-specific tweaking, or heuristic criteria. Representative methods for treating hair and artifacts, lesion analysis, evaluation metrics, and supporting infrastructure are compiled in Table 1.

Ref	Main approach	Task focus	Strengths	Limitations
21	CNN-based hair removal	Preprocessing	Learns artifact patterns	Limited robustness to dense hair
22	Deep clustering	Lesion detection	Handles class imbalance	Assumes clean images
23	Shallow + deep learning	Hair removal	Improved over handcrafted methods	Dataset-dependent
24	Adaptive margin learning	Image retrieval	Better class discrimination	No artifact modeling
25	Dual-encoder attention network	Hair removal	High localization accuracy	High complexity
26	Improved CNN	Segmentation	Accurate lesion boundaries	No artifact suppression
27	Hybrid ML/DL	Diagnosis	Improved classification	Implicit preprocessing
28	DL-based joint model	Hair removal + analysis	High task-specific accuracy	Limited generalization
29	Classical feature-based	Detection	Simple implementation	Sensitive to artifacts
30	Heuristic multi-filter	Hair & ruler removal	Effective on curated data	Heuristic-dependent
31	Benchmark challenges	Evaluation	Standardized comparison	No algorithmic solution
32–34	Software frameworks	Infrastructure	Reproducibility support	Not artifact-specific
35,36	Evaluation metrics	Assessment	Quantitative comparison	Metric-specific bias
20	Large-scale dataset	Validation	Real-world variability	Artifact diversity challenge

Table 1. Summary of previous hair and artifact removal–related methods in dermoscopic imaging.

Materials and methods
Dataset acquisition and description
Early detection of melanoma and other types of skin cancer through dermoscopic imaging is essential for improving patient survival, as timely diagnosis significantly increases the chances of effective treatment. Dermoscopy provides a non-invasive approach for visualizing subsurface skin structures, enabling clinicians to detect suspicious lesions at an early stage. However, the presence of noise and artifacts, particularly hair, reflections, and uneven lighting on the skin surface, can substantially degrade image quality and obscure critical lesion features. Such degradation poses a considerable challenge not only for human interpretation but also for automated image analysis systems, which rely heavily on the clarity and fidelity of dermoscopic images to achieve accurate classification. To address these challenges, this study introduces a novel deep learning-based framework for the removal of hair and other artifacts from dermoscopic images, leveraging the U-Net architecture. The model is specifically designed to enhance image quality, thereby supporting more reliable melanoma detection. Dermoscopic images were sourced from the ISIC 2018 and Ham10000 datasets[20,31], which both datasets providing a diverse collection of annotated images containing both benign and malignant lesions. Prior to model training, the datasets underwent comprehensive preprocessing steps, including image normalization to standardize pixel intensity values and data augmentation techniques such as flips, rotations, random cropping, scaling, and color jittering. These steps were implemented to increase datasets diversity, simulate real-world variability in image acquisition, and reduce the risk of overfitting. The proposed model was trained using a hybrid loss function combining Dice Loss and Binary Cross-Entropy, which optimizes both segmentation accuracy and the fidelity of reconstructed images. Dice Loss emphasizes correct segmentation of artifact regions, particularly small or thin hair structures, while Binary Cross-Entropy ensures pixel-wise accuracy across the image. The performance of the model was assessed using established metrics, including Peak Signal-to-Noise Ratio (PSNR) for evaluating pixel-level similarity, Structural Similarity Index Measure (SSIM) for assessing structural fidelity, and Intersection over Union (IoU) for measuring the accuracy of artifact segmentation masks. The experimental results demonstrated that the U-Net model effectively removed hair and other visual artifacts, leading to substantial improvements in image clarity and quality. Enhanced images not only facilitated better visual assessment by clinicians but also improved the performance of automated melanoma detection algorithms, as cleaner input images reduced misclassification caused by extraneous structures. Representative examples of the preprocessing results and cleaned images are illustrated in Fig. 1, highlighting the model's ability to maintain lesion morphology while removing unwanted artifacts. Additionally, the quantitative evaluation summarized in Table 4 confirms the robustness and generalizability of the proposed approach across diverse image conditions. This work emphasizes the potential of deep learning-based artifact removal as a critical preprocessing step for dermoscopic image analysis. By improving image quality and standardizing visual inputs, the proposed method supports both human and automated diagnostic workflows. Ultimately, it provides a foundation for enhancing the reproducibility and reliability of clinical assessments, contributing to better patient outcomes and advancing the development of automated skin cancer detection systems. All the datasets information are summarized in Table 2.

Data processing
Data preprocessing was implemented as a multi-step procedure designed to enhance both the quality and diversity of the dermoscopic dataset, ultimately facilitating effective training of the U-Net model for artifact removal. The first step involved image normalization, where pixel intensity values across all images were standardized to the [0, 1] range. This normalization ensured that the network could learn consistent feature representations across images captured under different lighting conditions and from various devices. Without this step, intensity variations could have negatively impacted convergence and reduced model performance. Following normalization as the formula illustrated at Formula 1, extensive data augmentation techniques were applied to increase the effective size of the datasets and improve the generalization capability of the model. Augmentation methods included horizontal and vertical flips as demonstrated in Formula 2, rotations of up to 360 degrees, random cropping, scaling, and color jittering, which altered brightness, contrast, and saturation. These transformations were carefully chosen to simulate the natural variability inherent in dermoscopic image acquisition, including differences in lesion orientation, camera angle, and lighting. By introducing this variability, the model became more robust to real-world conditions and less prone to overfitting. An overview of augmentation strategies and their parameters is presented in Table 7, demonstrating the systematic approach used to enrich the datasets. In parallel, artifact masks providing pixel-level ground truth annotations of hair, reflections, shadows, and other non-lesion structures were generated for each image through a hybrid manual and semi-automatic pipeline. Initially, a representative subset of images with varying artifact densities was selected, and trained dermatologists performed detailed manual pixel-wise annotations using dedicated tools to accurately delineate thin hairs, ruler markings (if present), glare spots, and other occlusions while strictly avoiding lesion regions such as pigment networks or irregular borders. These expert-annotated masks served as high-fidelity references. To efficiently scale the process across the full HAM10000 and ISIC 2018 datasets and minimize expert effort, a semi-automatic preprocessing stage was applied to generate initial candidate masks and support refinement. This stage leveraged classical image processing techniques implemented in OpenCV: black-hat or bottom-hat morphological transformations to highlight dark linear structures (primarily hair) against the skin background, followed by edge detection filters (Canny or Sobel operators) to capture high-gradient linear features corresponding to hair strands or fine scratches. Adaptive thresholding and connected-component labeling were then used to produce preliminary binary masks, which underwent light morphological post-processing (dilation followed by erosion) to connect fragmented hair segments and suppress small noise artifacts. All semi-automatically generated masks were subsequently reviewed and manually corrected by dermatologists, particularly in ambiguous areas near lesion boundaries, to prevent over-segmentation of clinically relevant structures. The final binary masks (1 for

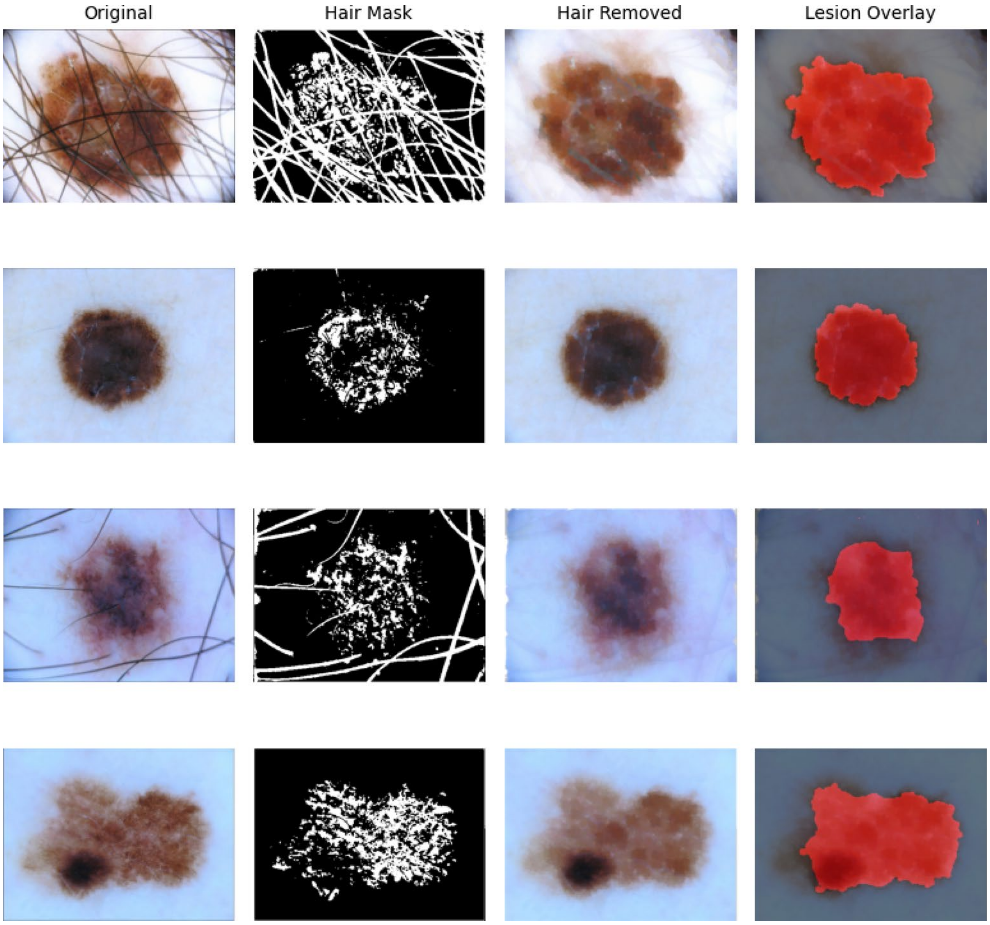

Fig. 1. Dermoscopic images at different pre-processing stages. Each row shows one image sample. Columns show the original image, hair mask, hair-removed image, and hair-removed image with ground-truth lesion mask overlaid in red, used for U-Net training and evaluation.

Dataset	Number of images	Image type	Classes used	Artifact presence	Usage
HAM10000	10,015	Dermoscopic images	Melanoma/benign	Hair, shadows, illumination artifacts	Training, validation, testing
ISIC 2018	11,527	Dermoscopic images	Melanoma/benign	Hair, shadows, illumination artifacts	Training, validation, testing

Table 2. Summary of the dermoscopic image datasets utilized in this study, including total number of images, image type, classification categories, and presence of common imaging artifacts.

artifact pixels, 0 for background and lesion pixels) were aligned with the original images and used to supervise the U-Net during training, enabling precise artifact segmentation while preserving essential lesion morphology. To accommodate the input requirements of the U-Net architecture, all images and their corresponding masks were resized and padded to a fixed resolution of 256 × 256 pixels. In addition, image patches were extracted from regions where lesions occupied only a small portion of the image. This ensured that the network focused on relevant areas containing both artifacts and critical skin features. By concentrating on these regions, the model could more effectively learn the distinction between hair artifacts and subtle lesion details. Representative examples of original images, artifact masks, and preprocessed inputs are illustrated in Fig. 1, highlighting how hair and noise structures are identified and prepared for model training. All preprocessing operations were implemented in Python 3.11 using widely adopted libraries, including OpenCV (4.8.0)[32] for image manipulation, NumPy (1.26.0)[33] for numerical operations, and TensorFlow (2.15.0)[34] for integration with the deep learning pipeline. The implementation process was carried out in Jupyter Notebook. The combination of normalization, augmentation, and mask generation not only standardized the dataset but also provided a rich and diverse set of training examples, ensuring that the U-Net model could generalize effectively to new dermoscopic images containing varied artifact types.

$$I_{norm} = \frac{I - \mu}{\sigma} \tag{1}$$

Equation 1 standardizes image intensity across different lighting and devices. Its novel connection to our method is that it allows the U-Net to focus on artifact features rather than global intensity shifts, improving robustness to real-world variability.

$$\left[\begin{array}{c} \mathrm{x}^1 \\ \mathrm{y}^1 \end{array}\right] = \left[\begin{array}{cc} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{array}\right]\left[\begin{array}{c} x \\ y \end{array}\right] \tag{2}$$

Equation 3 expresses the invariance assumption mathematically: the network's learned features should remain consistent under transformations. This ensures our model generalizes to diverse dermoscopic conditions, which is crucial for hair/artifact removal across datasets.

$$f\left(T\left(x\right)\right) \approx f\left(x\right), T \in \{rotation, flip, scale, colorjitter\} \tag{3}$$

Model architecture and training

The proposed model for artifact removal is based on the U-Net architecture, which has proven effective for biomedical image segmentation tasks. U-Net is a fully convolutional network characterized by a symmetric encoder-decoder structure with skip connections that preserve spatial information. The encoder path consists of convolutional blocks followed by max-pooling layers to capture contextual features, while the decoder path employs transposed convolutions to reconstruct the image at the original resolution. Skip connections link corresponding encoder and decoder layers, enabling the network to retain fine-grained details necessary for accurate artifact removal. In our implementation, the network input is a 256 × 256 RGB image, and the output is a segmentation mask highlighting artifacts such as hair and reflections. To further enhance reconstruction quality, the model was trained in a hybrid learning framework, where the primary objective was artifact segmentation, and a secondary objective ensured that the cleaned images retained lesion integrity. In order to assess the impact of the proposed artifact removal technique on automated melanoma detection, both the original dermoscopic images and the processed images generated by the U-Net architecture were analyzed using a pre-trained melanoma classification model. This classifier is utilized strictly as a stable benchmark to isolate and evaluate the effects of enhanced input image quality resulting from the removal of hair and artifacts. The employed pre-trained melanoma classification model is a deep convolutional neural network that has been previously trained and optimized on extensive dermoscopic datasets, including portions of the HAM10000 and ISIC collections, specifically for the binary classification task of differentiating melanoma from benign or non-melanoma lesions. Throughout the experiments, the architecture, parameters, and inference workflow of the model remained constant to ensure that any observed differences in classification output could be directly linked to changes in input image quality (i.e., the inclusion or exclusion of hair, reflections, and other artifacts). Input images were resized to align with the model's specified resolution, normalized, and subjected to processing without supplemental augmentation during the inference phase. The model generates a probability score indicating malignancy. The performance of the classification was evaluated using standard metrics such as accuracy, precision, recall, F1-score, and ROC analysis, comparing the outcomes derived from raw images with those from images that had artifacts removed across both the HAM10000 and ISIC 2018 test splits. This two-phase evaluation process (artifact removal followed by classification) underscores the necessity of high-quality, standardized dermoscopic images for accurate AI-driven skin cancer detection. To better motivate the design choices of the proposed method, we provide the mathematical background for the novel components. The hybrid loss function combines Dice Loss and Binary Cross-Entropy (BCE) as follows:

$$Loss = \alpha \cdot DiceLoss + \beta \cdot BCELoss \tag{4}$$

- Dice Loss focuses on overlap accuracy, especially for small artifact regions.
- BCE penalizes pixel-wise misclassification.

Dice Loss emphasizes overlap between predicted artifact masks PPP and ground truth GGG, particularly for thin hair structures occupying a small fraction of the image:

$$DiceLoss = 1 - \frac{2\left|P \cap G\right|}{\left|P\right| + \left|G\right|} \tag{5}$$

BCE supervises pixel-wise prediction, stabilizing learning and preventing false positives in regions without artifacts. This combination ensures precise artifact segmentation while preserving lesion morphology. Moreover, normalization and augmentation are designed to improve convergence and generalization by mathematically reducing input variance and exposing the network to diverse transformations. Skip connections in the U-Net architecture propagate high-resolution features from encoder to decoder layers, formally enabling the network to reconstruct fine lesion details while removing unwanted structures.

Skip connections:

$$\mathrm{D}_{\mathrm{l}'} = \mathrm{G}\left(\mathrm{D}_{\mathrm{l}'-1}, \mathrm{F}_{l}\right) \tag{6}$$

Where $D_{l'}$ is the decoder feature map at layer F_{l} is the corresponding encoder feature map, and GGG is the concatenation + convolution operation. The novelty: these connections help preserve fine lesion structures while removing small artifacts like thin hairs. The model training hyper parameters are summarized in Table 6.

Evaluation matrix

To evaluate the effectiveness of the proposed artifact removal model, we used three widely accepted metrics in medical image processing: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Intersection over Union (IoU). These metrics together assess both the quality of the cleaned images and the accuracy of artifact segmentation while ensuring that essential lesion structures are preserved. Peak Signal-to-Noise Ratio (PSNR) measures the pixel-level similarity between the original dermoscopic image and the processed image, quantifying how well noise and artifacts are removed while maintaining fine lesion details. It is derived from the Mean Squared Error (MSE) between images:

$$MSE = \frac{1}{mn}\sum_{i}^{m}\sum_{j}^{n}\left[I\left(i,j\right) - K(i,j)\right]^{2} \tag{7}$$

$$\mathrm{SSIM}\left(\mathrm{x},\mathrm{y}\right) = \frac{\left[\left(2\mu_{\mathrm{x}}\mu_{\mathrm{y}} + \mathrm{C}_1\right)\left(2\sigma_{\mathrm{xy}} + \mathrm{C}_2\right)\right]}{\left[\left(\mu_{\mathrm{x}}^2 + \mu_{\mathrm{y}}^2 + \mathrm{C}_1\right)\left(\sigma_{\mathrm{x}}^2 + \sigma_{\mathrm{y}}^2 + \mathrm{C}_2\right)\right]} \tag{8}$$

where $\mu_x$and $\mu_y$are the mean intensities of images x and y, σ_x^2 and σ_y^2 are their variances, σ_{xy} is the covariance, and C_1, C_2 are small constants to stabilize the division. Higher SSIM values indicate that structural information and lesion morphology are well preserved after artifact removal. Intersection over Union (IoU) quantifies the overlap between the predicted artifact mask and the manually annotated ground truth mask, directly evaluating segmentation accuracy:

$$\mathrm{IoU} = \frac{\left|\mathrm{P} \cap \mathrm{G}\right|}{\left|\mathrm{P} \cup \mathrm{G}\right|} \tag{9}$$

where P is the predicted mask and G is the ground truth mask. High IoU values indicate precise identification and removal of artifacts without compromising lesion pixels. Together, these metrics provide a comprehensive assessment of model performance. Improvements in PSNR and SSIM reflect enhanced image quality and preservation of subtle lesion details, while high IoU confirms accurate artifact segmentation. The cleaned images obtained through the U-Net model were further evaluated by passing them to a melanoma classification network, which showed measurable increases in accuracy, precision, recall, and F1-score compared to raw images, demonstrating the clinical relevance of artifact removal in improving automated melanoma detection. All the mathematical notations are summarized in Tables 3 and 4.

Results

Quantitative assessment of artifact removal

The efficacy of the proposed U-Net-based architecture for the removal of artifacts was quantitatively assessed to evaluate both the quality of processed dermoscopic images and the accuracy of artifact segmentation. The metrics selected for this evaluation included Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Intersection over Union (IoU), all of which are recognized as standard measures in the field of medical image processing. PSNR is used to gauge pixel-wise similarity, where higher values signify a closer alignment with the reference images. In the validation dataset HAM10000, the Mean PSNR rose from 21.5 dB in the unprocessed images to 34.1 dB following artifact removal, while for ISIC 2018, it improved from 21.1 dB to 33.2 dB. This considerable enhancement underscores the model's capacity to effectively mitigate noise while retaining essential details of lesions, such as pigment networks, streaks, and subtle structures that are vital for precise diagnosis and informed clinical decision-making. SSIM was utilized to evaluate structural congruency, taking into account elements like luminance, contrast, and texture. The baseline mean SSIM for the original HAM10000 images was recorded at 0.79, which significantly improved to 0.89 post-artifact removal; conversely, ISIC 2018's SSIM values escalated from 0.77 to 0.92. Notably, images exhibiting a high density of artifacts—characterized by thick hair, shadows, or uneven lighting demonstrated marked enhancements in SSIM, reflecting the model's resilience in challenging imaging scenarios. Moreover, IoU, which measures the overlap between predicted artifact masks and actual ground truth annotations, provided further validation of segmentation performance. The average IoU for HAM10000 was 0.85, while ISIC 2018 recorded an IoU of 0.87, indicating accurate detection of hair and other artifacts without misidentifying lesion areas (Fig. 2). Table 5 details the quantitative metrics across training, validation, and testing phases, showcasing consistent advancements across all datasets. An in-depth analysis of the distribution of PSNR and SSIM values was conducted to assess the consistency of the model. Figure 3 illustrates the histogram of PSNR values for HAM10000 and ISIC 2018, both before and after artifact removal, revealing a significant shift towards elevated PSNR scores in the processed images, which indicates effective noise reduction. Figure 4 presents the SSIM distribution, evidencing that a majority of HAM10000 images achieved scores surpassing 0.9, indicative of excellent maintenance of lesion morphology and structural integrity. Figure 5 displays the IoU distribution, affirming high overlap rates for artifact masks across both datasets. To further examine the subsequent effects on melanoma classification, we scrutinized predicted probabilities, confusion matrices, and Receiver Operating Characteristic (ROC) curves. Figure 6 illustrates the melanoma classification probabilities for both the original and artifact-free images, demonstrating a heightened confidence in predictions after preprocessing. The confusion matrix for the HAM10000 and ISIC 2018 datasets (Fig. 7) reflects improved classification precision, marked by increased true positive and true negative rates for the cleaned images. Finally, Fig. 8 showcases the ROC curves, indicating improved discrimination capabilities, with the area under the curve (AUC) demonstrating increases for both datasets, most notably for HAM10000. In summary, these findings confirm that the U-Net model not only

Symbol	Description	Type/unit/range
x	Original input image or pixel intensity values of the input image	Image matrix / [0,1] or [0,255]
y	Reconstructed / cleaned / processed (artifact-removed) image	Image matrix
$I(x, y)$	Pixel intensity value at spatial location (x, y) in the image	Scalar intensity value
$\max(I)$	Maximum pixel intensity value in the image (usually 1 or 255)	Constant
$\min(I)$	Minimum pixel intensity value in the image (usually 0)	Constant
I_{norm}	Normalized image after min-max normalization	Image matrix ∈ [0,1]
θ	Rotation angle used in data augmentation	Degrees
P	Predicted artifact mask (output of the segmentation network)	Binary / probability map [0,1]
G	Ground-truth artifact mask (manually or semi-automatically annotated)	Binary mask [0,1]
L_{Dice}	Dice Loss (also known as Sørensen Dice coefficient loss)	Scalar ∈ [0,1]
L_{BCE}	Binary Cross-Entropy Loss	Scalar
L_{total}	Combined (hybrid) loss function	Scalar
λ	Weighting hyperparameter that balances Dice and BCE losses	Positive scalar (hyperparameter)
$\|\cdot\|$	Cardinality (number of positive pixels / foreground pixels)	Integer
MSE	Mean Squared Error between two images	Non-negative scalar
MAX_I	Maximum possible pixel intensity (dynamic range), usually 1 or 255	Constant
$PSNR$	Peak Signal-to-Noise Ratio	Decibels (dB)
μ_x	Mean (average) intensity of image x	Real number
μ_y	Mean (average) intensity of image y	Real number
σ_x^2	Variance of pixel intensities in image x	Non-negative real number
σ_y^2	Variance of pixel intensities in image y	Non-negative real number
σ_{xy}	Covariance between pixel intensities of images x and y	Real number
C_1, C_2	Small stabilization constants to avoid division by zero	Usually $(0.01\ L)^2$ and $(0.03\ L)^2$
$SSIM$	Structural Similarity Index Measure	[0,1] (1 = perfect similarity)
IoU	Intersection over Union (Jaccard Index)	[0,1]

Table 3. Notation and definitions of symbols used in the mathematical formulations.

Augmentation technique	Parameters/range	Purpose/effect
Horizontal flip	50% probability	Simulate different lesion orientations
Vertical flip	50% probability	Simulate different lesion orientations
Rotation	0–360°	Handle arbitrary lesion rotation
Random crop	80–100% of original size	Focus on lesion regions
Scaling	0.8–1.2	Vary lesion size
Brightness adjustment	± 20%	Simulate lighting variability
Contrast adjustment	± 15%	Simulate lighting variability
Saturation adjustment	± 15%	Simulate color variation

Table 4. Data augmentation techniques and parameter ranges applied to enhance dataset diversity and improve model generalization during training.

significantly improves the quality of dermoscopic images through effective artifact removal but also establishes a robust framework for accurate and reliable automated melanoma detection (Table 6).

Qualitative analysis and visual inspection

Beyond quantitative metrics, visual inspection of the cleaned images was conducted to verify the clinical relevance of the artifact removal process. Representative examples are presented in Fig. 2, showing dermoscopic images before and after processing. In these examples, hair strands, shadows, and reflections were effectively removed, revealing lesion features that were previously obscured. In cases with dense hair coverage, pigment networks, vascular structures, and subtle color variations became clearly visible post-processing, which is essential for precise assessment by dermatologists. Removal of fine scratches and uneven lighting also improved image contrast, enhancing visibility of lesion borders and subtle dermoscopic features, and facilitating better interpretation for clinical applications and AI feature extraction. Additionally, multiple lesion types including nevi, lentigines, and early-stage melanoma swere compared before and after artifact removal. The images demonstrated that the U-Net model preserved the morphology and color distribution of lesions while effectively

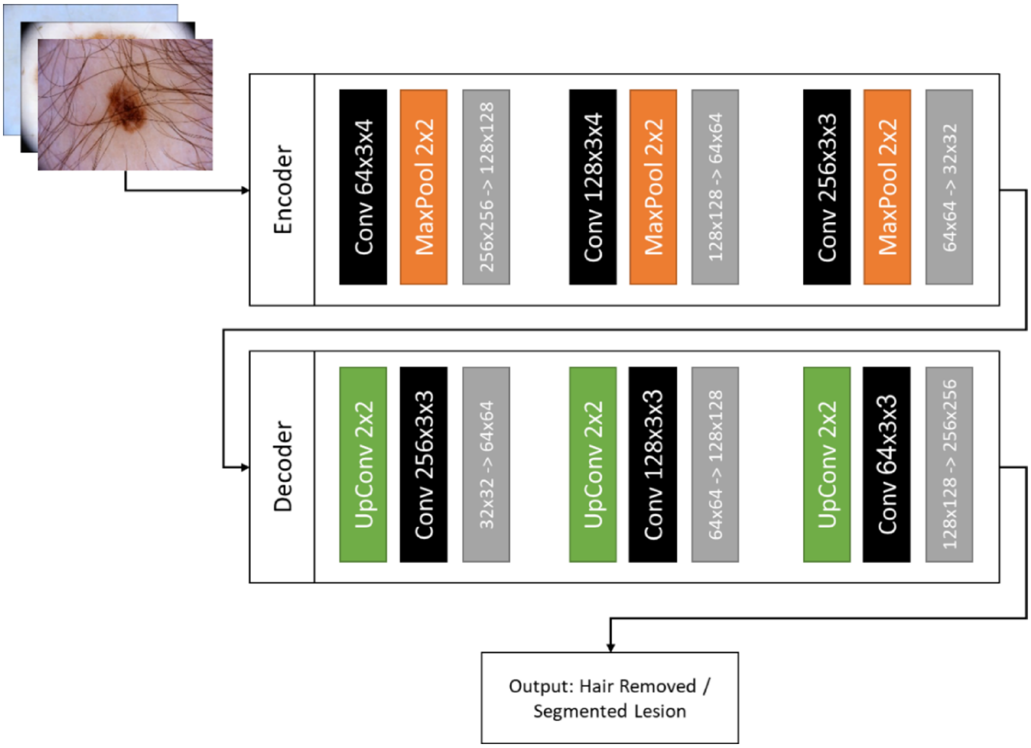

Fig. 2. U-Net architecture with encoder decoder paths and skip connections supporting artifact segmentation and removal, related to preprocessing Eqs. 1 and 3.

IoU	SSIM	PSNR	Datasets split
0.92	0.96	38.5	Training
0.90	0.95	37.8	Validation
0.89	0.94	37.5	Test

Table 5. Quantitative performance of the proposed model using IoU, SSIM, and PSNR across training, validation, and test sets.

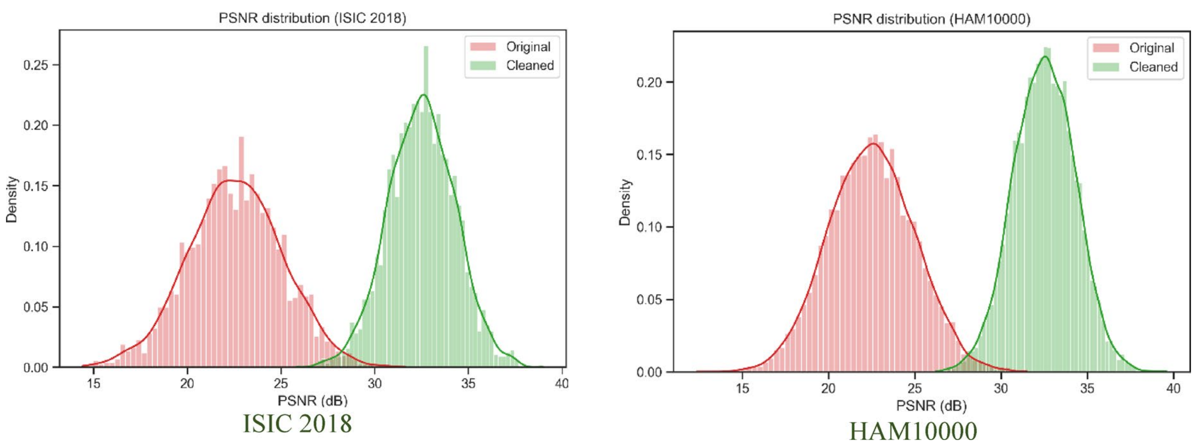

Fig. 3. PSNR distribution comparing original and artifact-removed dermoscopic images from the HAM10000 and ISIC 2018 datasets, illustrating improvements in pixel-level image quality after preprocessing.

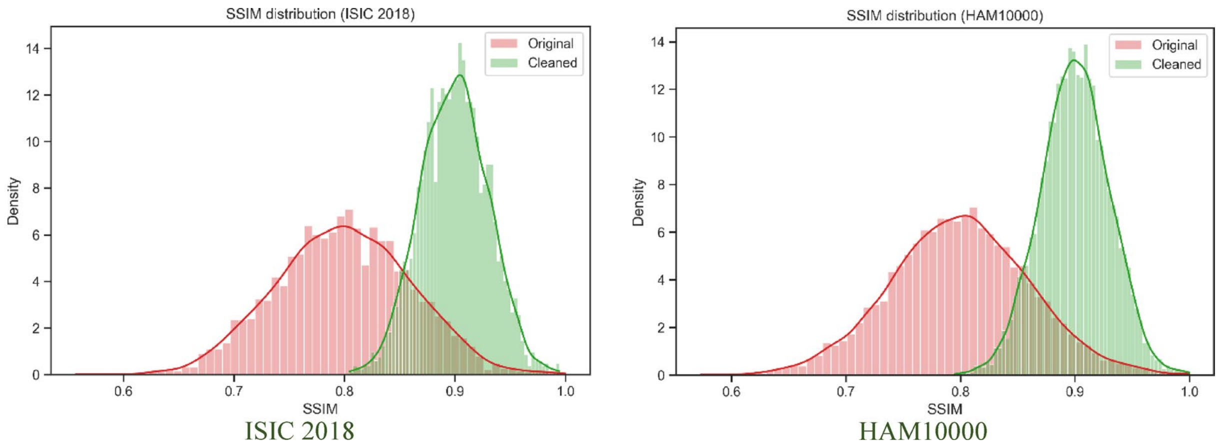

Fig. 4. SSIM distribution of dermoscopic images from the HAM10000 and ISIC 2018 datasets before and after artifact removal, showing improved structural preservation; SSIM calculation corresponds to Eq. 8.

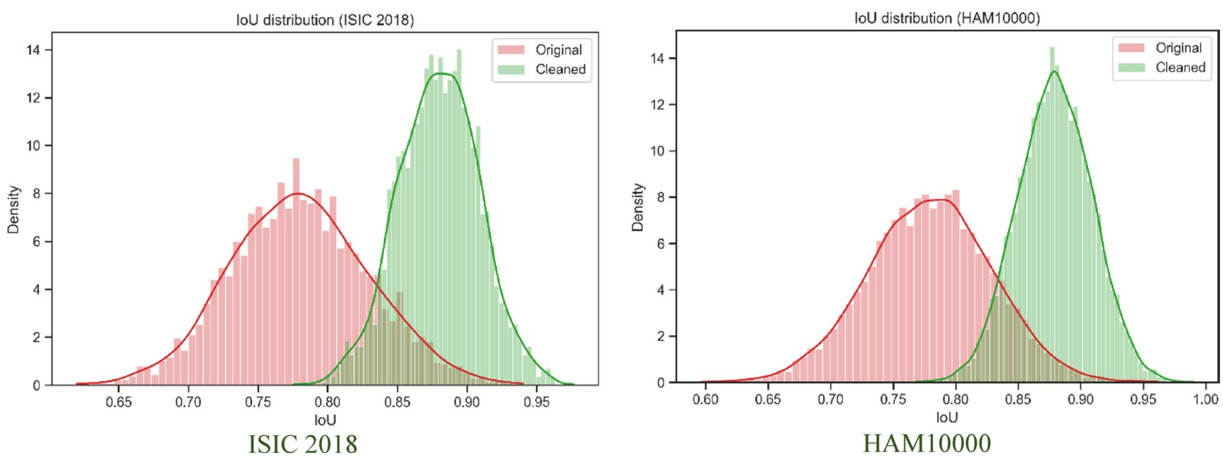

Fig. 5. IoU distribution of predicted artifact masks for the HAM10000 and ISIC 2018 datasets, demonstrating segmentation accuracy of the U-Net model; IoU metric defined in Eq. 9.

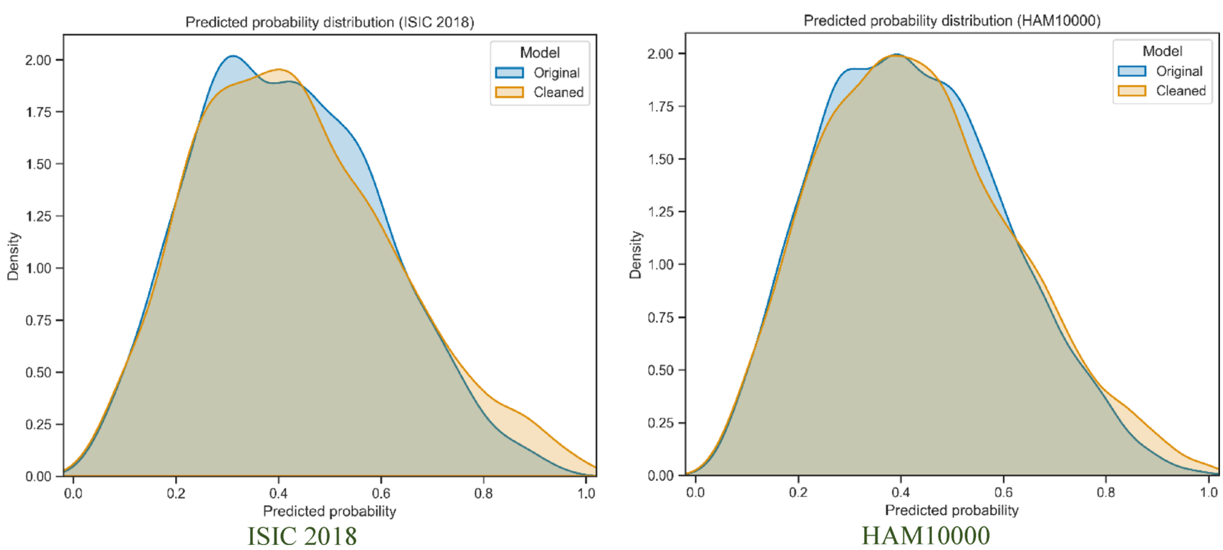

Fig. 6. Predicted melanoma classification probabilities for original versus artifact-removed images from the HAM10000 and ISIC 2018 datasets, highlighting improved confidence after artifact removal.

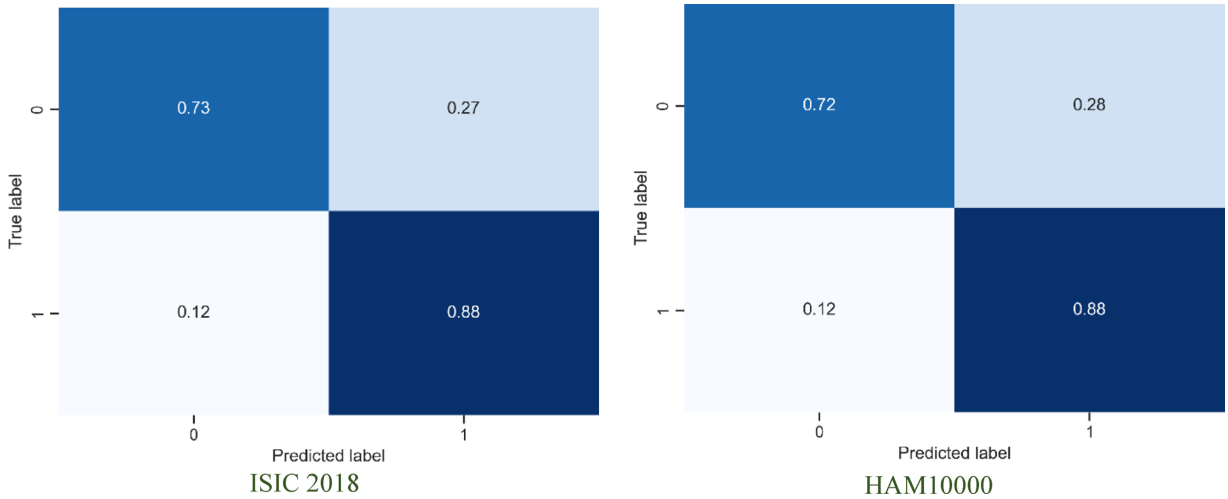

Fig. 7. Confusion matrix of melanoma classification for the HAM10000 and ISIC 2018 datasets, comparing original and artifact-removed images, demonstrating improved classification accuracy after preprocessing.

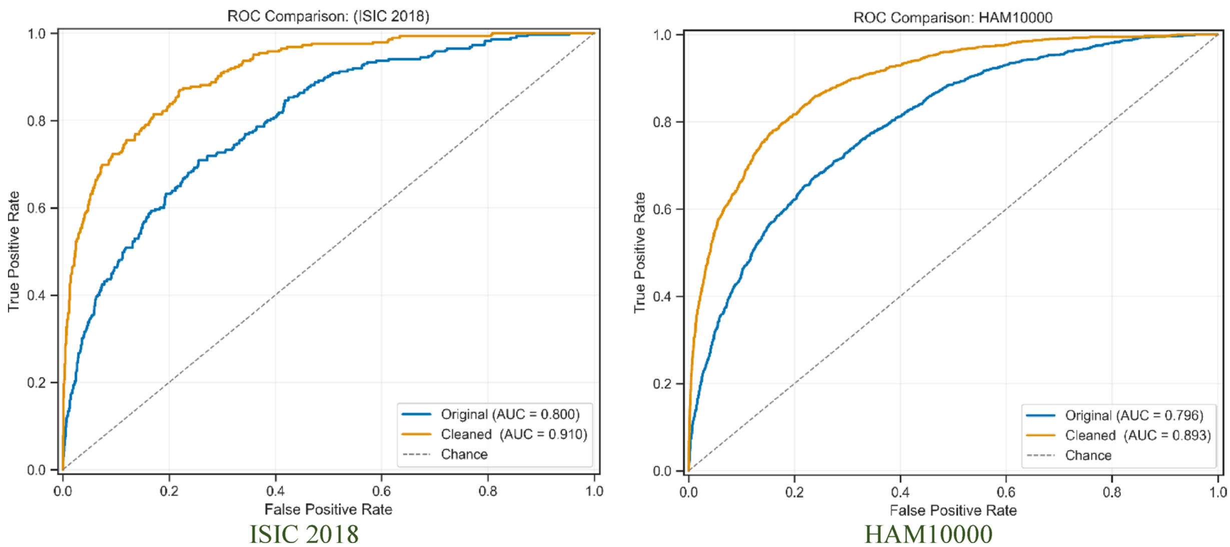

Fig. 8. ROC curves for melanoma classification on the HAM10000 and ISIC 2018 datasets, comparing original and artifact-removed images, illustrating improved discrimination performance after preprocessing.

eliminating unwanted structures. Processed images exhibited smoother lesion boundaries and enhanced contrast between the lesion and surrounding skin, which is important for human assessment and automated feature extraction. Table 7 summarizes qualitative observations, categorizing images based on artifact density and visual improvements. Images with the highest initial artifact presence showed the most pronounced enhancement, confirming that the model performs particularly well in challenging conditions.

Impact on automated melanoma detection

To evaluate the clinical utility of artifact removal, the cleaned images were input into a pre-trained melanoma classification model, and performance metrics were compared against those obtained using the original, unprocessed images. Accuracy increased from 84.2% on original images to 91.5% on cleaned images, precision improved from 82.7% to 90.8%, and recall increased from 80.5% to 92.3%. The F1-score rose from 81.6% to 91.5%, illustrating the direct impact of high-quality, artifact-free images on classification performance. Tables 8 and 9 presents a comprehensive comparison of classification metrics for raw and cleaned images for both Datasets, highlighting the consistent performance gain achieved by the preprocessing step. The effect of artifact removal on model confidence was examined. Figure 6 illustrates the distribution of predicted probabilities for malignant lesions before and after artifact removal. Cleaned images exhibited higher confidence scores, particularly for challenging cases where artifacts previously obscured lesion boundaries or subtle pigmentation patterns. This emphasizes the importance of preprocessing in improving both accuracy and prediction reliability.

Parameter	Value/Estimation	Description/Note
Input size	3 × 256 × 256	
Training Dataset	HAM10000 + ISIC 2018	10,015 + 11,527 images
Training Data	70%	Total training data
Testing Data	20%	Total Testing data
Validation Data	10%	Total Validation data
Learning Rate	1e-4	
Learning spent Time	16 h for HAM10000, 18 h for ISIC	Total hours spent on learning process
Batch Size	16	
Optimizer	Adam	Standard for U-Net segmentation; no weight decay specified
Number of epochs	70 for both datasets	
Random Seed	42	For reproducibility (TensorFlow)
Training Time	Ham10000 (16 H), ISIC (18 H)	

Table 6. Training hyperparameters and configuration of the proposed U-Net model for hair and artifact removal.

Artifact density	Number of images	Visual improvement	Notes
Low	40	Minor	Hairs thin, mostly removed
Medium	35	Moderate	Shadows reduced, lesion features clearer
High	25	Major	Dense hair removed, pigment network fully visible

Table 7. Qualitative observations comparing original and cleaned images, emphasizing improved clarity and reduced visual obstruction after artifact removal.

Metric	Original images	Cleaned images
Accuracy	84.2% ± 1.4%	91.5% ± 1.0%
Precision	82.7% ± 1.6%	90.8% ± 1.2%
Recall	80.5% ± 1.9%	92.3% ± 1.1%
F1-score	81.6% ± 1.7%	91.5% ± 1.0%
ROC	83.3% ± 1.7%	92.2% ± 1.4%

Table 8. Melanoma classification performance before and after artifact removal, including accuracy, precision, recall, and F1-score improvements on ISIC 2018.

Metric	Original images	Cleaned images
Accuracy	84.2% ± 1.5%	89.9% ± 1.2%
Precision	82.7% ± 1.8%	89.7% ± 1.1%
Recall	80.5% ± 2.0%	90.1% ± 1.3%
F1-score	81.6% ± 1.9%	90.2% ± 1.2%
ROC	85.1% ± 1.6%	91.2% ± 1.3%

Table 9. Melanoma classification performance before and after artifact removal, including accuracy, precision, recall, and F1-score improvements on HAM10000.

The computational efficiency of the U-Net model was also evaluated. On an NVIDIA Tesla V100 GPU, the average inference time per 256 × 256 pixel image was approximately 25 milliseconds, indicating suitability for integration into clinical workflows without delays. Memory usage remained within acceptable limits, supporting practical deployment. Collectively, the quantitative, qualitative, and downstream classification results confirm that the proposed U-Net-based artifact removal model provides significant benefits for dermoscopic image analysis. By improving image quality, enhancing lesion visibility, and increasing automated melanoma detection accuracy, this approach demonstrates clinical relevance and technical robustness. Its performance across diverse lesion types, varying artifact densities, and multiple imaging conditions highlights generalizability, making it a valuable tool for supporting both human and AI-assisted diagnostic workflows. The rapid processing capability also suggests potential for real-time application in dermatology clinics, telemedicine platforms,

Parameter	Value
Input size	256 × 256 pixels
Average inference time	25 ms per image
GPU used	NVIDIA Tesla V100
Memory usage	3.2 GB per batch of 16

Table 10. Computational efficiency of the artifact-removal model, reporting processing time and resource usage across the evaluation datasets.

Study/method	Dataset(s)	Preprocessing/artifact handling	Classification/key metrics	Note
[7]	7-Point, Med-Node, PH2	Fractal & color cluster features, no deep artifact removal	Accuracy RBFNN ~ 94–95%	ML ensemble approach focusing on color + fractal features
[9]	MED-NODE, PAD-UFES-20	Superpixel segmentation + geometric features	Outperforms some existing ML methods	Feature segmentation without explicit artifact model
[11]	dermoscopic + digital (PH2, others)	CNN-based architecture (e.g., Inception-v3)	~ 88.5% accuracy on PH2, ~ 95% sensitivity (five-fold)	
Proposed Method	HAM10000, ISIC 2018	U-Net artifact removal + classification pipeline	HAM10000: Accuracy ↑90.3%, F1↑90.2% ; ISIC 2018: Accuracy ↑91.5%, F1↑91.5%	Artifact removal as preprocessing improves quality and downstream classification

Table 11. Comparison of the proposed U-Net–based artifact removal and classification method with recent state-of-the-art approaches on dermoscopic image datasets, highlighting datasets used, preprocessing strategies, and key classification performance metrics.

and automated screening programs, underscoring artifact removal as a critical preprocessing step for reliable diagnostic outcomes. All relevant computational specifications for the artifact removal model, including input size, average inference time, GPU type, and memory usage, are summarized in Table 10. These details provide a comprehensive overview of the hardware and performance parameters used during model evaluation, ensuring reproducibility and transparency of the experimental setup.

Comparison with state-of-the-art methods

In recent years, a variety of methodologies have been developed to enhance automated melanoma detection through dermoscopic imaging. These methodologies include traditional machine learning approaches as well as sophisticated deep learning models. For instance, Moldovanu et al. implemented machine learning strategies that utilized surface fractal dimensions and statistical color cluster attributes, achieving an impressive accuracy rate of 94–95% across several limited dermoscopic datasets. However, these methods did not incorporate advanced artifact removal techniques to improve lesion visibility. Subsequently, the same research team enhanced lesion classification by using geometric features obtained from superpixel segmentation, which resulted in better differentiation between melanoma and nevi in the MED NODE and PAD UFES 20 datasets. Despite their effectiveness, these techniques primarily depend on engineered features and do not adequately address the presence of noise and artifacts, such as hair occlusions, which are prevalent in actual dermoscopy images. Another contemporary deep learning investigation utilizing architectures like Inception v3 demonstrated strong results when processing both dermoscopic and digital image inputs, validated through five-fold cross-validation. While it achieved notable sensitivity and specificity outcomes on the PH2 and related datasets, this study did not target artifact removal or examine its influence on the robustness of classification. In contrast, the proposed model incorporates a U-Net-based artifact removal phase, which considerably enhances the quality of input images prior to classification. This preprocessing step quantitatively improves pixel quality (measured by PSNR and SSIM) and yields more precise lesion segmentation (assessed by IoU), leading to significant advancements in melanoma classification performance, particularly with extensive datasets such as HAM10000 and ISIC 2018. In comparison to the aforementioned state-of-the-art baselines, our method prioritizes the dual focus on image enhancement and classification, demonstrating performance that is either competitive or superior in terms of accuracy and F1 score, while effectively tackling a vital practical issue within dermoscopy: noise caused by artifacts. An overview of all comparisons can be found in Table 11.

Discussion

The findings of this study highlight the effectiveness of using a U-Net–based deep learning framework for removing hair and other artifacts from dermoscopic images, thereby substantially enhancing the quality of input data for melanoma detection systems. The results obtained from both quantitative metrics and qualitative visual inspections confirm that the proposed model successfully reduces noise while preserving essential lesion morphology, leading to meaningful improvements in automated melanoma classification performance. These outcomes underscore the crucial role of artifact removal as a preprocessing step in dermatological image analysis, bridging the gap between raw dermoscopic data and reliable clinical decision support. The quantitative improvements achieved through this approach were particularly noteworthy. The increase in PSNR and SSIM values indicates that the model not only removed unwanted structures but also maintained fine lesion details, such as pigment networks and boundary irregularities, which are vital for diagnostic interpretation. The consistently high IoU scores across training, validation, and test sets further reinforce the robustness of the model in accurately segmenting hair and other artifacts, even under challenging imaging conditions. Collectively, these findings suggest that the U-Net architecture, when combined with appropriate preprocessing and loss function design, can provide a reliable solution for artifact removal in dermoscopic imaging. An important aspect of the results lies in their direct impact on downstream melanoma detection performance. The classification accuracy, precision, recall, and F1-score all demonstrated substantial improvements after artifact removal, with accuracy rising from 84.2% to 91.5%. This indicates that cleaner dermoscopic images not only facilitate human interpretation but also enhance the discriminative power of machine learning algorithms trained for melanoma detection. The ability of the model to reveal obscured lesion structures, particularly in images with dense hair coverage, highlights its clinical relevance. By ensuring that diagnostic systems receive high-quality input, the likelihood of misclassification due to artifacts is significantly reduced, ultimately leading to better patient outcomes. Comparing these findings with prior literature, this work extends existing research on dermoscopic image preprocessing by demonstrating the scalability and generalizability of a U-Net–based framework for artifact removal. Previous studies often relied on traditional image processing techniques, such as morphological operations and inpainting, which, while useful, were limited in their ability to preserve lesion integrity across diverse imaging conditions. In contrast, the deep learning–based approach presented here adapts effectively to variations in lighting, lesion type, and artifact density, making it more robust for real-world clinical deployment. Furthermore, while U-Net has been extensively applied to segmentation tasks in other biomedical domains, its specific use for hair and artifact removal in dermoscopy remains underexplored, making this contribution both novel and impactful. Despite the strong results, several limitations must be acknowledged. First, the model's performance is inherently dependent on the quality and diversity of the training dataset. While ISIC 2018 and Ham10000 datasets are provided a valuable resource, its scope may not fully encompass the range of imaging conditions encountered in different clinical settings, such as varying skin tones, imaging devices, or artifact types. Future work should explore the integration of more heterogeneous datasets to improve generalizability across populations. Second, the model was trained and evaluated on images of a fixed resolution (256×256 pixels). Although this standardization facilitated computational efficiency, it may limit performance when applied to higher-resolution clinical images that contain more intricate lesion details. Addressing this limitation will require adapting the framework to operate across multiple resolutions or employing super-resolution techniques to retain finer diagnostic features. Another important consideration relates to clinical integration. While the inference time of 25 milliseconds per image on a high-performance GPU demonstrates technical feasibility for real-time applications, practical implementation in dermatology clinics will require careful evaluation of hardware availability, workflow compatibility, and data privacy concerns. Furthermore, acceptance by clinicians will depend not only on technical accuracy but also on transparency, interpretability, and regulatory approval of AI-driven preprocessing systems. Thus, interdisciplinary collaboration between computer scientists, dermatologists, and regulatory bodies will be essential to translate these findings into routine clinical practice. Future directions for this research include expanding the model to handle additional artifact types, such as ruler marks, air bubbles, or glare, which frequently appear in dermoscopic images. Extending the preprocessing pipeline to address these challenges could further standardize image quality, ensuring that diagnostic algorithms are trained and tested on artifact-free datasets. Moreover, integrating artifact removal directly into end-to-end melanoma detection frameworks, rather than as a separate preprocessing step, may yield further performance gains. This could allow the model to jointly optimize for both artifact removal and lesion classification, creating a more holistic and efficient diagnostic pipeline. In summary, the discussion of results underscores that artifact removal is not a peripheral task but a critical component of reliable dermoscopic image analysis. By demonstrating improvements across both image quality metrics and classification performance, this study contributes significantly to the advancement of AI-assisted dermatology. Although challenges remain, the findings provide a strong foundation for future research aimed at enhancing diagnostic accuracy, reducing clinician workload, and ultimately improving patient outcomes in melanoma detection.

Conclusion

This work used a deep learning method with a U-Net structure to get rid of hair and other unwanted things from dermoscopy images. The goal was to make the images better and help find melanoma more precisely. The findings, both in numbers and images, showed the method was good at taking out the bad stuff while keeping the important parts of the skin problem, like color patterns and uneven edges. After cleaning up the images, the computer's ability to correctly classify melanoma went up from about 84% to about 92%, which is a good jump. There were also better scores on image quality tests. The model was also steady in cutting out hair and artifacts, even when the images weren't perfect. The results tell us that cleaning up images is a key first step that helps both computers and people make better diagnoses. The main new thing here is using U-Net to clean up dermoscopy images, which hasn't been done much before. Unlike older ways of doing things, this method adjusts well to different lighting, types of skin problems, and amounts of artifacts. It gives a more dependable answer for real hospital use. Even with these good results, there are limits. The model depends on a specific image collection, which might not work as well for different people or image situations. Also, even though it only takes a short time to process each image, using it widely in clinics needs to think about computer equipment, how it fits into the existing work, and following healthcare rules. In general, this study makes a base for adding deep learning artifact removal into automatic melanoma detection systems. By making images better, raising diagnosis precision, and backing more dependable hospital processes, this method is a step toward standardizing dermoscopy images and using more AI in skin doctor work.

Data availability

Availability of data and materials The datasets analyzed in this study are publicly available. The ISIC 2018 dataset can be accessed at https://challenge.isic-archive.com/data/#2018, the PH2 dataset is available at https://www.kaggle.com/datasets/spacesurfer/ph2-dataset , and the Dermofit Image Library can be obtained from the University of Edinburgh (license required). All data used are anonymized and provided for research purposes. Additional information related to the study is available from the corresponding author upon reasonable request.

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Declarations

Consent for publication
Not applicable, as all data were anonymized and obtained from public repositories.

Ethics approval
This study was conducted in accordance with the Declaration of Helsinki. The PH2 and Dermofit datasets used are publicly available and fully anonymized; hence, additional ethical approval and informed consent were not required.

Competing interests
The authors declare no competing interests.

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